

Discussion of "Profit Sharing in Critical Minerals Value Chains: Evidence from a Direct Lithium Revenue-Sharing Agreement with Indigenous Communities"

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Motivation

- **A personal connection:** I lived for many years in Tierra Amarilla (Atacama), a mining-intensive municipality.
- My town faced **local costs of mining:** blasting vibrations, dust, and air pollution.
- How can communities bearing these costs also **share in the benefits of extraction?**



Tierra Amarilla, Atacama

This paper studies a unique institutional arrangement in Chile: direct revenue sharing with Indigenous communities.

A very interesting paper!

- **An important question:** how should resource rents be shared with communities affected by extraction?
- **A new setting:** private–public–community relationships in Chile’s lithium industry.
- **Thorough empirical evidence:** educational outcomes at the individual level and economic activity at the municipality level.

My comments focus on the nature of the transfers, identification, and the underlying mechanisms.

1. What exactly is the transfer?

- **Albemarle transfers 3.5% of sales:** 3.0 p.p. for community development and 0.5 p.p. for monitoring, research, and CPA management.
- **Economically substantial.** If it was a transfer, roughly **US\$3,300–3,900 annually for a family of three**
- **Mainly used for community projects:** water and sanitation, community facilities, and scholarships.
 - Yet **limited public information** on the allocation of expenditure.
- **Other transfers to the SPA municipality:** Albemarle (about US\$1.2m, 2017–2020) and SQM (about US\$27m, 2018–2022).
 - Interesting, but challenging, to contrast these transfers with the community ones

Which projects drive the results?

Household support? Infrastructure? Productive activities and local procurement?

2. Control group: info-robustness-extensions

- **Mining municipalities are heterogeneous.** Similar pre-treatment characteristics do not imply similar exposure to subsequent sectoral shocks.
 - E.g., copper-intensive municipalities versus lithium-intensive San Pedro de Atacama.
 - Heterogeneous growth in tertiary education centers across municipalities (reform in 2016)
- **Potentially useful comparison: Antofagasta.** Albemarle's La Negra downstream processing facilities are located there, outside SPA.
 - Exposure to Albemarle's expansion, without the direct CPA agreement.
 - Could provide **complementary evidence** on production-related spillovers.
- **More information on synthetic-control weights W_m :** donor composition, pre-treatment fit, and robustness to alternative donor pools.

3. Mechanisms in tertiary education results

- **Financial constraints are a plausible mechanism.** But why are the effects particularly strong among vulnerable students?
- **Chile introduced free tertiary tuition (*gratuidad*) for vulnerable students in 2016 (universities) and 2017 (technical institutes).** The paper controls for it, but tuition may not be the main remaining constraint for these students.
- **Could food and housing support be particularly important?**
 - How were scholarships implemented? Cash, reimbursements, or in-kind support?
 - **Do students attend institutions outside SPA** in the Antofagasta/Atacama/Coquimbo regions?
- **Can we also examine tertiary enrollment/retention?** More recent high-school cohorts could provide additional evidence on the educational effects of the agreement, even before degree completion can be observed.

Useful additional evidence: Scholarship eligibility and recipients; enrollment, retention rates, and institution location.

4. Identification and mechanisms: municipality level results

- **2016 was not an isolated transfer shock.** Albemarle also renegotiated larger extraction quotas, followed by increased production.
- **Mining expansion may generate local spillovers** even if most mining workers live outside SPA (approximately 25% local, according to the draft).
 - Local procurement, transport, construction, and other services may expand with production.
- **Tourism is another major source of local activity.** Was its growth independent of the transfers, or did community investments complement tourism?

Can we distinguish exposure to the mining expansion from exposure to the CPA transfers? Sectoral composition might help

- **Mining-linked sectors:** potentially production-driven spillovers.
- **Community-project-linked sectors:** potentially transfer-driven local demand.

Final thoughts

- **An important and timely paper!** Direct revenue sharing offers a potentially effective way for communities to benefit from resource extraction.
- **The educational results are particularly interesting**, especially the larger effects among vulnerable students.
- **More evidence on the allocation of funds and sectoral spillovers** could further strengthen the interpretation of the economic effects.